Hardware Acceleration for Cluster Fault Tolerance in Hybrid CMOS/non-CMOS Memories

Dissertation Defence

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1. Motivation and Problem

Why this project matters

- Hybrid CMOS/non-CMOS memories offer high density and low cost, but they are vulnerable to **spatially correlated cluster faults**.
- Traditional ECC schemes are often evaluated for **independent random errors**, not realistic burst faults.
- Previous studies mainly relied on simulation; **hardware-validated comparison** was still missing.

Core research question

Which ECC design gives the best trade-off between fault tolerance, latency, area, power, and storage efficiency on real FPGA hardware?

2. Project Goal and Objectives

Project goal

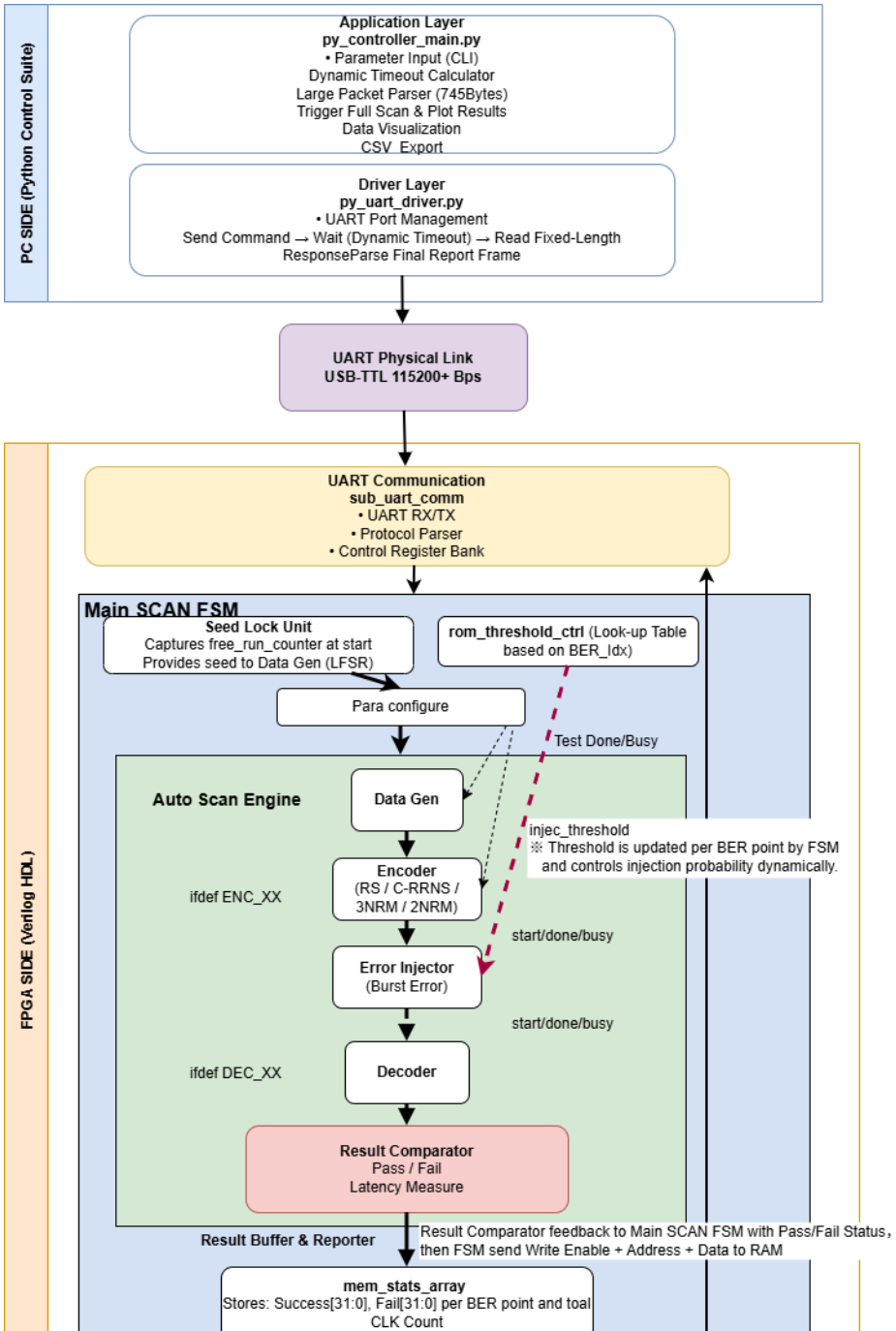
To design and implement an **FPGA-based hardware acceleration platform** for fair benchmarking of multiple ECC algorithms under realistic cluster-fault conditions.

Objectives

1. Implement and compare **6 configurations**:
   - 2NRM-RRNS Parallel
   - 2NRM-RRNS Serial
   - 3NRM-RRNS
   - C-RRNS-MLD
   - C-RRNS-MRC
   - RS(12,4)
2. Evaluate four key dimensions:
   - fault tolerance
   - decoding latency
   - FPGA resource and power cost
   - storage efficiency # 3. Methodology and Experimental Setup

Evaluation flow

4. FPGA Platform Architecture



5. Main Innovation: Probabilistic Fault Injection Engine

Key features

- Uses a **32-bit Galois LFSR** as the pseudo-random source.
- Supports both **random single-bit** and **cluster burst** injection.
- BER is controlled by threshold comparison instead of a fixed schedule.
- Boundary-safe burst masks are precomputed offline and stored in ROM.

Why it matters

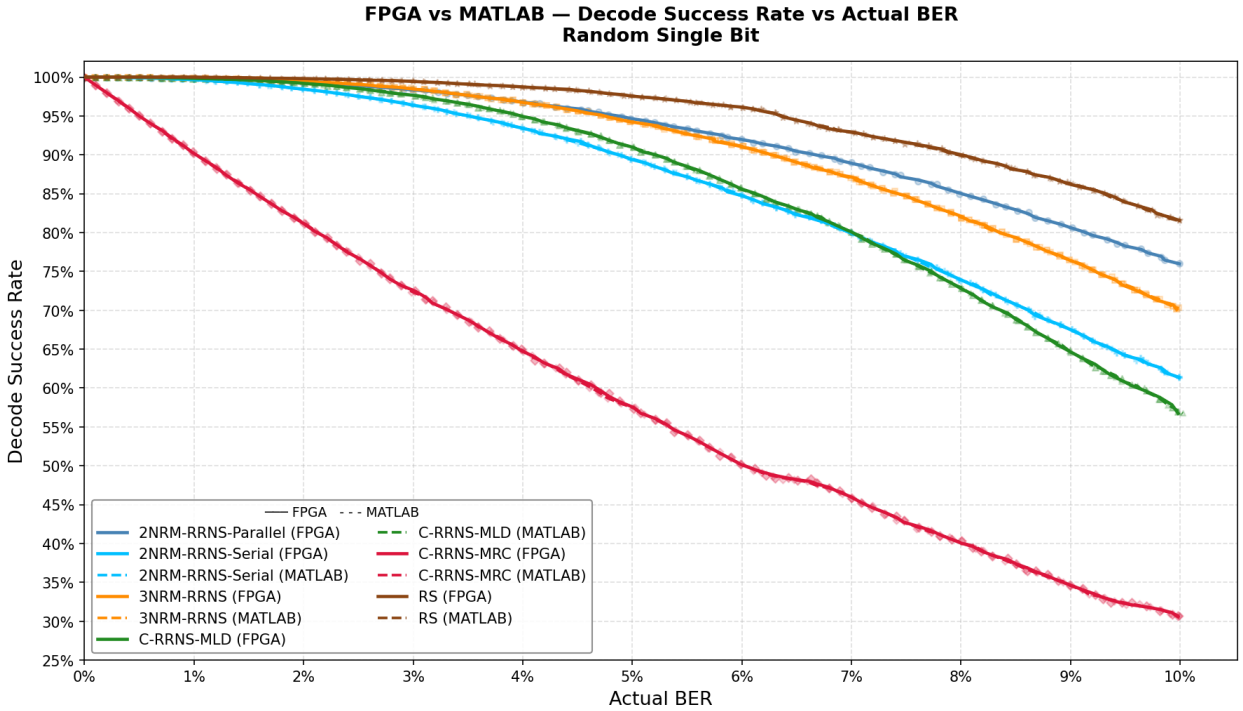
This makes the platform **resource-efficient, reproducible, and reusable** for benchmarking different ECC algorithms.

6. Why the Test Platform Is Strong

Advantages of the proposed FPGA test platform

- **Fair comparison**: all algorithms share the same test infrastructure and clock condition.
- **Hardware validated**: results are measured on a Xilinx Artix-7 FPGA, not simulation alone.
- **Low overhead fault injection**: the probabilistic injector uses only **two BRAMs**.
- **Scalable experiments**: supports **101 BER points** from 0% to 10% and **100,000 samples per point**.
- **Extensible architecture**: new ECCs can be added with the wrapper interface.
- **Cross-validated**: FPGA curves closely match MATLAB results, increasing credibility.

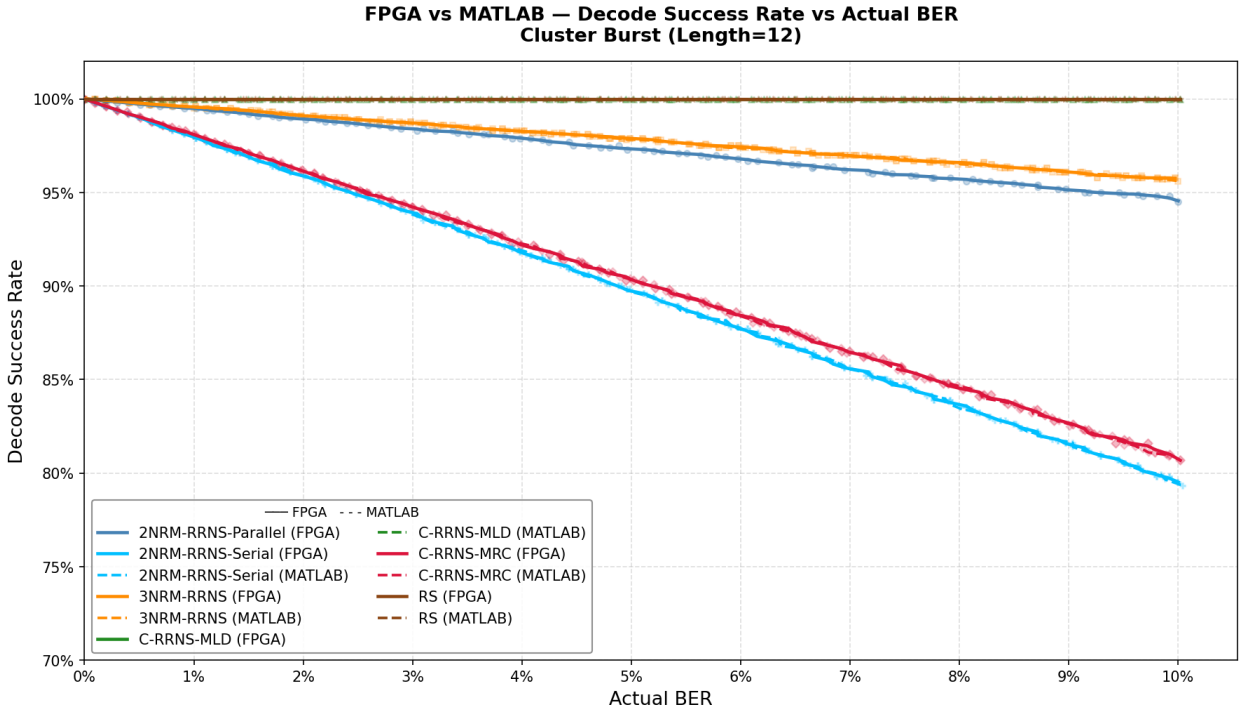
7. Random Single-Bit Results



Main observation

- FPGA and MATLAB curves show **close agreement**.
- Correcting ECCs remain strong at low BER, while **C-RRNS-MRC** drops fastest.

8. Cluster Burst Results at Representative Length L=12



Main observation

- **C-RRNS-MLD** and **RS(12,4)** show the strongest burst resilience.
- Burst-fault ranking differs from random-bit testing, showing why a realistic fault model is necessary.

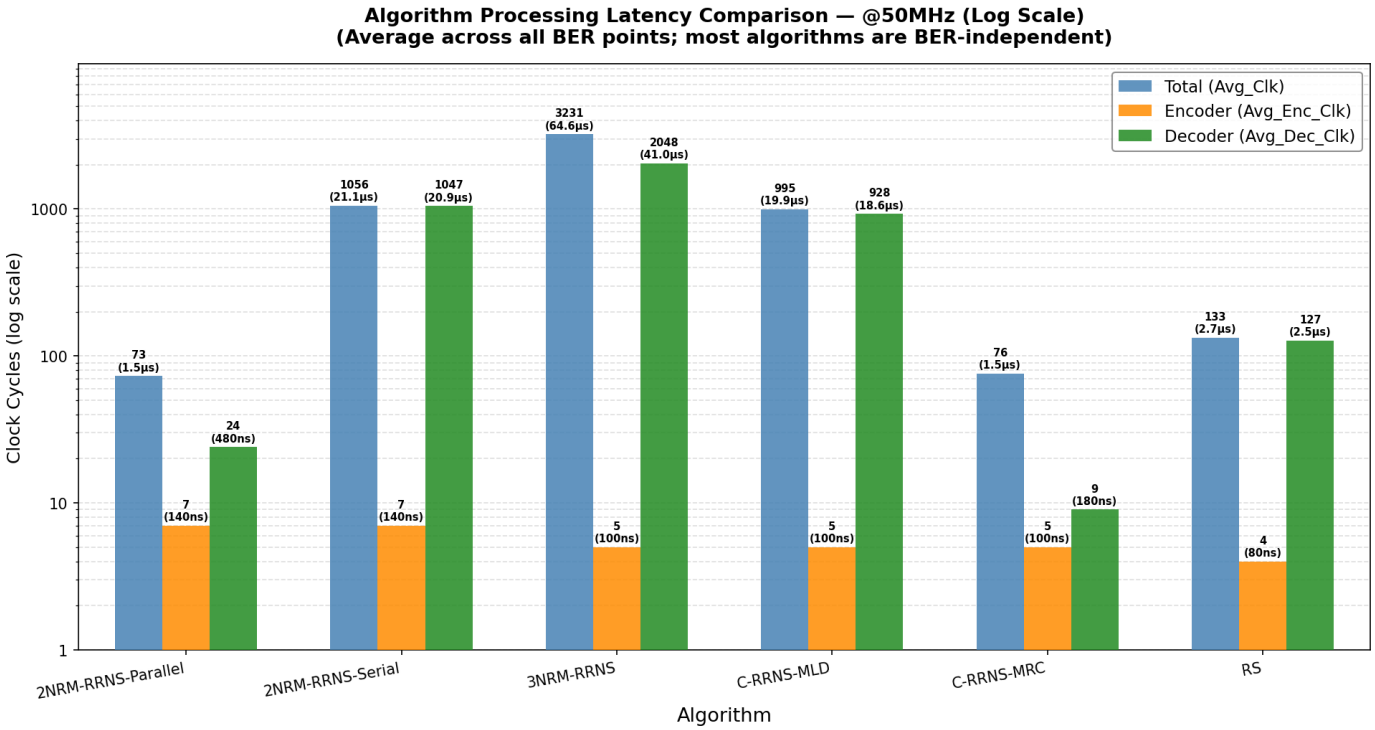
9. Quantitative Comparison of Key Results

Algo	Burst L_{max}	Dec. cycles	Efficiency
C-RRNS-MLD	**14**	928	26.2%
RS(12,4)	13	127	33.3%
3NRM-RRNS	11	2048	33.3%
2NRM-P	8	**24**	**39.0%**
2NRM-S	7	1047	**39.0%**

Takeaway

- **Most reliable**: C-RRNS-MLD
- **Fastest**: 2NRM-P
- **Most compact**: 2NRM

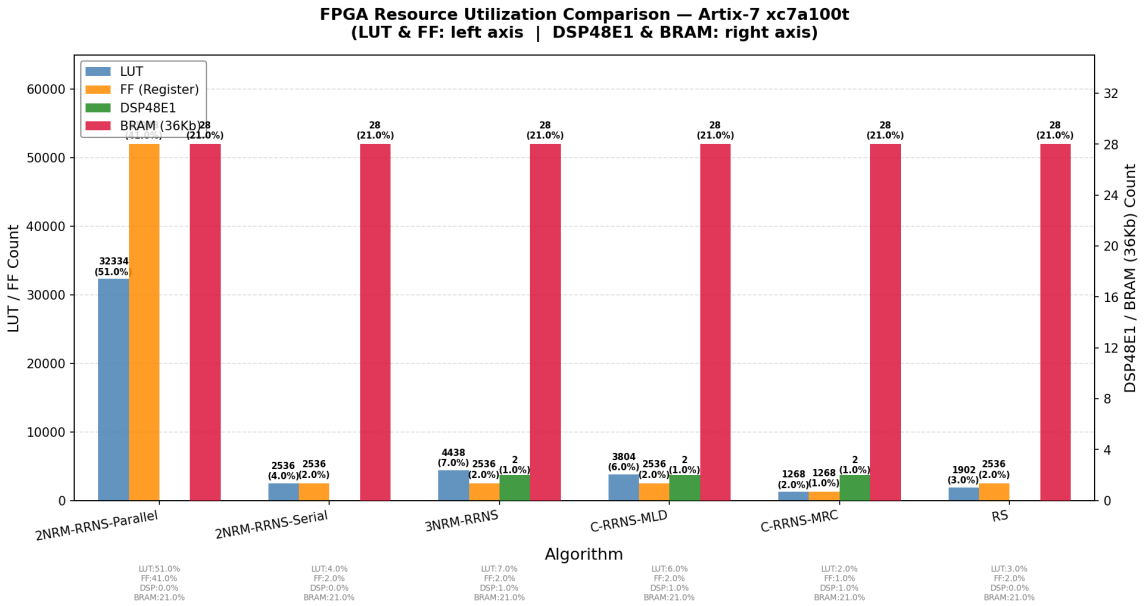
10. Latency Comparison



Conclusion from latency results

- The **2NRM-RRNS Parallel** decoder is the fastest correcting design.
- The parallel implementation achieves a major acceleration over the serial version.
- The result clearly exposes the **resource-latency trade-off** of parallel MLD.

11. Resource and Storage Comparison



Key message

- **2NRM-Parallel** achieves high speed at the highest LUT/FF and power cost.
- **2NRM** is the most storage-efficient option, while **C-RRNS-MLD** gives the strongest protection.

12. Main Conclusions

Major conclusions from the dissertation

1. The project successfully delivers a **working FPGA benchmarking platform** for ECC evaluation under cluster faults.
2. **C-RRNS-MLD** shows the strongest burst-fault tolerance in the explored design space.
3. **2NRM-RRNS Parallel** offers the lowest decode latency, but its area and power cost are high.
4. **2NRM-RRNS** provides the best storage efficiency for overhead-sensitive systems.
5. The close agreement between FPGA and MATLAB results confirms the **correctness and reliability** of the evaluation method.

13. Contributions and Future Work

Contributions

- A reusable FPGA evaluation platform with unified wrappers.
- A low-resource probabilistic fault injection engine.
- The first hardware-level quantification of the **parallel vs serial 2NRM-RRNS trade-off**.

Future work

- evaluate wider data words and more ECC variants
- reduce LFSR correlation with stronger random sources
- migrate the framework toward ASIC or real hybrid memory devices

Appendix A. Fault Injection Algorithm

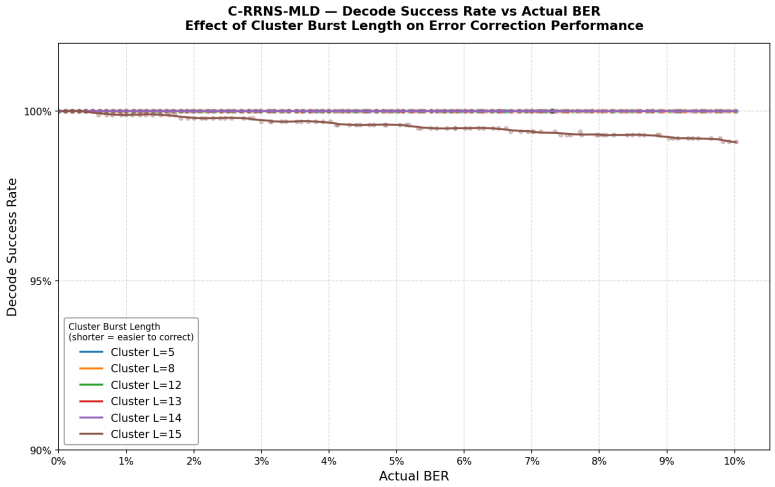
Probabilistic injection procedure

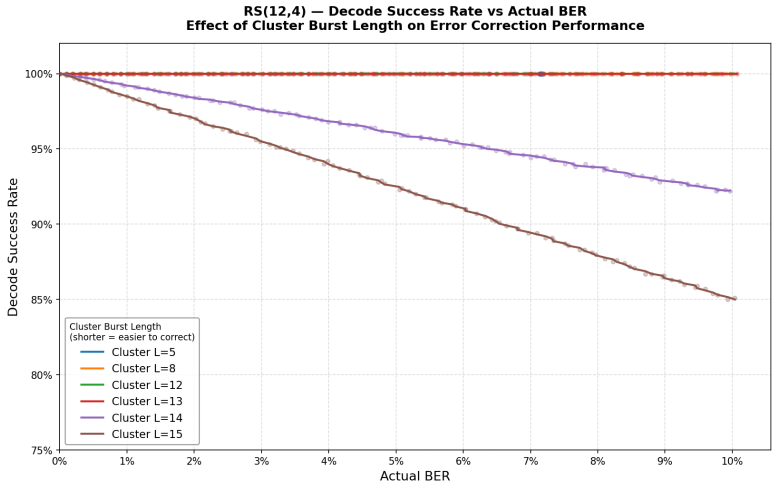
1. Generate encoded codeword.
2. Lock one random seed for the full BER sweep.
3. For each BER point, compute the injection threshold.
4. Use the 32-bit Galois LFSR to decide whether to inject an error.
5. If random mode is selected, test valid bits independently.
6. If burst mode is selected, apply one ROM-based contiguous burst mask.
7. Decode and compare with the original data to record pass or fail.

Purpose

This appendix shows that the fault injector is not a simple random flip unit, but a **controlled and reproducible hardware evaluation algorithm**.

Appendix B. Backup Figure: Burst-Length Impact





Backup message

These curves support the final claim that **C-RRNS-MLD** provides the best extreme cluster-fault tolerance, with **RS** as the strongest balanced alternative.

Thank You

Questions?

Thank you for listening.